

10 Papers: Cancer Drug & Oncology Trials

PACLITAXEL BIOSYNTHESIS AI BREAKTHROUGH

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ABSTRACT

Paclitaxel, $C_{45}H_{51}NO_6$, biosynthesis is an active area of research due to ongoing progress towards more sustainable and environmentally friendly production of the drug compound. Recent literature details the characterization of enzymes that play a role in synthesis, optimization of growth media, and RNA related regulatory mechanisms. The method of PhD students spending excessive time performing literature reviews to discover new findings is obsolete due to faster and high quality state of the art conversational AI. In this study, representative AI times were obtained regarding how long would it take the fastest human researcher to read, analyze, extract information, and type a high quality 250 word answer, with the fastest time of 1380 seconds being used as a standard reference. The slowest AI generation in the study was 79.19s by ChatGPT-4o, which was still over 17x faster than the optimal human performance time. Here, a paclitaxel biosynthesis breakthrough was illustrated using LLMs and LLMs. In the first instance, full length papers were summarized by AI models – with the finding that AI provided more detailed answers across entire papers, generating over 10x longer descriptions and 12x faster times compared to the manuscript author's methods to summarize abstracts. The outputs of individual AI generated answers yielded a 10 Paper Summary with 6,322 words, and served as the input for eight separate prompts, which provided valuable insights regarding both emerging and historical views of paclitaxel biosynthesis, engineering microorganisms, as well as top 10 new research recommendations, and top 10 strategies for this area. The second instance, biosynthesis breakthrough was demonstrated with a speed of 732 seconds for 36 generations compared to the single optimal human response of 1,380 seconds. Top models received an average AI judge score of 4.5 by ChatGPT-4o for Part A, a score of 9.3 by o1-preview, L3.1-405B, and ChatGPT-4o for Part B, and a score of 8.3 by ChatGPT-4o and 5.5 Sonnet, followed by a score of 9.2 for GPT-4o for Part C. These superior results have primarily been afforded by OpenAI, ClaudeAI, and Meta AI new model releases in late 2024 that have helped to advance the paclitaxel biosynthesis field. The presence of speedups with more detailed answers over optimal human responses is supported by advanced clinical hardware that processes high dimensional and complex data continually to solve combinatorial problems such as those in this study using 15 different prompts across 163 generations.

Keywords: Paclitaxel, Biosynthesis, Artificial Intelligence

1 INTRODUCTION

With the establishment of a chromosome-level high-quality reference genome map of *Taxus* in 2021, the biosynthesis of anti-cancer drug Paclitaxel shown in Figure 1 has become a more active area of research due to the potential to increase drug compound yield using economical and convenient culture. Therefore, 10 2024 abstracts were summarized by the manuscript's author in the following paragraphs for comparison purposes to AI regarding the taxane drug's biosynthesis.

Li, C., et al. in May 2024 typically characterized a novel cyclohexane P450 enzyme called Oxetan synthase (TaxCP1) that efficiently catalyzes the formation of the oxetane ring in paclitaxel. Using 1H and ^{13}C labelling and density functional theory calculations, a possible reaction mechanism was proposed involving intramolecular oxidocyclization rearrangement through a concerted process published in [1].

As one of the most potent chemotherapeutic agents against a range of malignant tumors, the increasing demand for paclitaxel has led to the development of multiple strategies to address supply issues instead of extracting this compound from the bark of slow growing yew trees due to its low content in *Taxus plants*. Li, C., et al. 2024 [1].

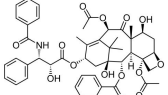


Figure 1: Paclitaxel Skeletal Structure, $C_{45}H_{51}NO_6$

Decker, A., et al. in August 2024 combined *in silico* methods and functional gene analysis to help characterize the taxane hydroxylase, which is needed to oxygenate paclitaxel intermediates. The characterization of a specific taxane hydroxylase, referred to as *TJH10* was identified as imperative for heterologous systems used to improve taxane yields. The authors also identified a potential missing enzyme referred to as TB574 that may permit further exploration of novel metabolic engineering approaches with microbial consortia shown in [2].

MONOCLONAL ANTIBODY BIOPROCESS ENGINEERING ADVANCEMENTS USING CONVERSATIONAL ARTIFICIAL INTELLIGENCE

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ABSTRACT

Processing high dimensional and complex monoclonal antibody (mAb) bioprocess data in industry is now more efficient due to conversational AI. The human in the loop approach to Large Language Model (LLM) interfacing with document and/or chained outputs is a valuable benefit to existing biotechnology workflows. Potential risks of using natural language processing are minimized due to the utility of solving problems with vast amounts of structured and unstructured mixed data that can be verified by the Human-AI team. This novel work demonstrates o1-preview, ChatGPT-4o, L3.1-405B, and 3.5 Sonnet models' best and state-of-the-art solutions. In specific, o1-preview provided a response to 19 papers 11x faster than the manuscript author's time after the number of words were set equal. In addition, ChatGPT-4o was 271x faster than an optimal human researcher to examine and provide an estimate regarding dimension reduction or combinatorial optimization for a recent paper by Kan, M., et al. The third LLM speed advantage of 3.06x by ChatGPT-4o vs. the manuscript author was achieved using movie carlo simulations and markov chain models performance forecasts and a current paper by Kowolik, P., et al.

Part A featured the individual analysis of 5 recent mAb production papers, which emphasized the proficiency of o1-preview (9.0/10), ChatGPT-4o (9.2), and L3.1-405B (9.2) providing a forecast report. Example generations for o1-preview and L3.1-405B typically established connections between using dimension reduction or combinatorial optimization and improving bioprocesses. Part B models generated tables regarding how LLMs can improve numerical data from 3 different papers using movie carlo simulations or markov chain models. An example from ChatGPT-4o (9.0) was substantially more complete, accurate, and convincing than the table provided by 3.5 Sonnet (8.0). Part C outlined the report format from Part A combined with the numerical approach from Part B across 6 additional papers, led by o1-preview (9.0) and ChatGPT-4o (8.5). The o1-preview example followed the prompt format well, citing each of how LLMs will utilize reinforcement learning and bayesian optimization to improve mAb production. The work represents a standard for utilizing a consumer amount of bioprocess data to forecast new results, with the transition into LLMs providing near-real-time production data analysis aided by document retrieval to provide a synergistic effect with existing machine learning techniques.

Keywords: Monoclonal Antibodies, Bioprocess Engineering, Conversational Artificial Intelligence

1 INTRODUCTION

Monoclonal antibodies (mAbs) are the active proteins representing many biologies and biotechnologies on the market in 2024. Several human conditions have been treated through the use of commercial mAbs such as cancer, osteoporosis, and COVID-19 [1, 2, 3]. Oncological drugs such as trastuzumab (2), bevacizumab (3), and rituximab have each seen manufacturing increases in recent years due to proven drug effectiveness and loss of patented biologic drug exclusivity [4].

As of 2020, FDA recommended a reliable and continuous source of antibody production based on master cell banks for cell lines or clones obtained from animals that have been immunized with the substance that is the subject of study [6]. The cell lines are produced by fusing cells from the immunized animal with myeloma cells, as reported by Köhler, G., et al. in 1975 [7], and illustrated in Figure 9.

Enzyme scale production of monoclonal antibodies has increased due to advances in culture medium concentrations, downstream processing methods becoming widely accepted, and the use of non-chemotherapeutic reagents demonstrated by Shukla, A., et al. [9]. Analogously, Kelley, B., found that the increase in mAb yields was due to an increase in titers to reduce



Figure 1: NISTmAb Reference Material Example [5]

mAb BIOPROCESS ENGINEERING IN-CONTEXT TABLE FORECASTS USING CONVERSATIONAL AI LITERATURE INSIGHT GENERATIONS

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ABSTRACT

Bioprocess engineering has incorporated effective AI applications in recent years that consist of traditional approaches to training models on relevant data to then analyze and predict new and unseen data. The missing component has been the ability to process mixed data from an assortment of dissimilar information sources with high contextual awareness to inform the Human-AI team on how LLMs and other authors' methods will further improve results. Here, real-time search and document retrieval was used to generate a mixed speed analysis of over 600 by 3.5 Sonnet were obtained vs. the manuscript author regarding monoclonal antibody (mAb) bioprocess engineering kinetics across several models. ChatGPT-4o with an average score of 9/10 was the leader in quality for this task with several detailed reports that were obtained using document search addressing a paper's specific weaknesses being improved with LLMs or two other author's methods. This protocol was applied systematically for each of the other two papers, supported by the other two relevant bioprocess papers. o1-preview's advanced reasoning as a new standard over five other models in processing either 136 extracellular or 101 intracellular metabolite tables, incorporating the analysis of 12 additional paper summaries across two prompts with two table revisions. For extracellular metabolites, o1-preview generated an 18 metabolite table including all metabolite forecasts that were expected to be breakthroughs due to future integration of a LLM or other author's recent methods. The model supported in forecasts with interpretable author citations and quotations for breakthrough metabolites, along with lists of author specific and metabolite specific insights that influenced its conclusions. For intracellular metabolites, o1-preview provided a full 101 metabolite table, matching the number of entries from the original Subramaniam, P., et al. table, including confirmation for each metabolite regarding whether each forecasted value was expected to be a breakthrough. Overall, this work was represented by numerous speed advantages, literature insights to address paper weaknesses, and competent o1-preview in-context table analysis with supporting evidence from leading articles represented by two 8,510 scores to lead the first conversational AI mAb bioprocess engineering revolution.

Keywords: Bioprocess Engineering, Monoclonal Antibodies, Table Forecasts, Literature Insights, Conversational AI

1 Introduction

Neural networks, support vector machines, and linear regression are some of the many types of AI incorporated into modern bioprocess engineering workflows [1]. AI models work by training on known experimental data so that when new data is introduced, the AI model can predict relevant outcomes or make additional decisions in production to high profitability. Artificial intelligence has also been used in "recombinant biologics production in host systems such as mammalian cells (CHO and HEK293T) [1, 2].

Additional studies have implemented AI into automation to improve process flexibility, enhance data reliability, and increase productivity. Extensive research using AI has also been applied to pharmaceutical and functional genomics within the field of biotechnology [3]. In addition, use modelling and data-driven insights have led to incorporation of AI with digital twins to obtain higher levels of manufacturing precision [4]. Machine learning has also been put into practice to "speed up the solving of physics-based models" regarding input-output relations in computational fluid dynamics [4].

These AI developments in bioprocess engineering have addressed data types that may lack diversity, and typically do not possess high contextual awareness to provide a solution to concurrent linguistic and numerical inquiries, providing AI with large amounts of tabulated data for inference while gaining insight from new reports with data to provide industry insights to mitigated to recent advances Large Language Models (LLMs) or Large Multimodal Models (LMMs) are capable of maintaining prior AI analytical performance, while using natural language processing (NLP) and supporting tools to read information and respond with intuition to researchers in English [5].

10 Paper source for green and sustainable Paclitaxel biosynthesis

- Results: o1-preview retrosynthesis. Sonnet: Insights.
- Advantages: May have influenced field by novel LLM use at scale.
- Limitations: Several decades of paclitaxel biosynthesis failures.

mAb bioprocess data LLM processing, state-of-the-art solutions by 3 LLMs

- Results: Forecasts: o1-preview 9.9, ChatGPT-4o 9.2, Llama 3.1-405B 9.2.
- Advantages: Multi-table settings based on 6 bioprocess papers.
- Limitations: o1-preview context length limited its broad high performance.

136 extracellular & 101 intracellular metabolite tables further studied

- Results: o1-preview scores = 9.5. Over 600x faster than author.
- Advantages: Competent in-context table analysis based on papers.
- Limitations: Insights based on a select table time interval of 4 days.

LLM Papers: Cancer Drug & Oncology Trials

CANCER VS. CONVERSATIONAL ARTIFICIAL INTELLIGENCE

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ABSTRACT

Solving cancer mechanisms is challenging due to the complexity of the disease integrated with many approaches that researchers take. In this study, information retrieval was performed on 40 oncological papers to obtain authors' methods regarding the tumor immune microenvironment (TIME) or organ-specific research. 20 TIME summaries were combined and analyzed to yield valuable insights regarding how research based papers complement information from review papers using Large Language Model (LLM) in-context comparisons, followed by code generation to illustrate each of the authors' methods in a knowledge graph. Next, the 20 combined organ-specific emerging papers impacting historical papers was obtained to serve as a source of data to update a mechanism by Zhang, Y., et al., which was further translated into code by the LLM. The new signaling pathway incorporated four additional authors' area of cancer research followed by the benefit they could have on the original Zhang, Y., et al. pathway. The 40 papers in the study represented over 600,000 words which were focused to specific areas totaling approximately 17,000 words represented by detailed and reproducible reports by Clao-3Opus. ChatGPT o1 provided advanced reasoning based on these authors' methods with extensive correlations and citations. Python or LaTeX code generated by ChatGPT o1 added methods to visualize Conversational AI findings to better understand the intricate nature of cancer research.

Keywords Cancer - Tumor Immune Microenvironment - Organ-Specific Cancer - Conversational AI

1 Introduction

LLMs have been increasingly utilized for cancer research in a number of applications. In April 2023, Naik, H., et al. utilized an earlier version of ChatGPT to generate a single patient case report regarding synchronous bilateral breast cancer [1]. In September of the same year, Choi, H., et al. collected data from reports of surgical pathology and ultrasounds from 2,931 breast cancer patients, and extracted the information using ChatGPT 3.5 to efficiently derive clinical factors, obtaining an overall accuracy of 87.7% [2]. On October 16, 2023, Griewing, S., et al. in *Journal of Personalized Medicine* compared the concordance of treatment recommendations from ChatGPT 3.5 with those of a multidisciplinary tumor board for breast cancer (MTB). The authors found that "Overall concordance between the LLM and MTB was reached for half of the patient profiles" [3].

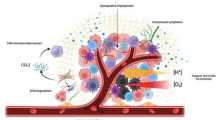


Figure 1: Tumor microenvironment, Garlisi, B., et al. 2024 [4]

Furthermore, in November 2023 Sorin, V., et al. utilized ChatGPT-3.5 or GPT-4 to process breast cancer clinical notes, question-answering based on guidelines, and patients' management recommendations. The rate of correct answers varied from 64-98%, with the highest accuracy (88-98%) observed in information extraction and question-answering tasks [5]. A study published on November 17, 2023 focused on utilizing a variety of LLMs to examine "10 fictional cases of patients with advanced cancer with genetic alterations. Each case was submitted to 4 different LLMs (ChatGPT, Galactica, Perplexity, and BioMedLM) and 1 expert physician". Results of 4 lung cancer cases and 6 other cancer cases indicated that the LLMs in precision oncology did not yet reach the quality and credibility of human experts [6].

40 paper source: TIME & Organ-specific tumors mechanisms

- Results: o1 Python TIME knowledge graph, LaTeX mechanisms.
- Advantages: 3 Opus LLM insight: 600K word papers to 17K words.
- Limitations: Generated visual code required author improvements.

CLINICAL DECISION SUPPORT BASED ON BEVACIZUMAB CANCER TRIALS AND PUSHING THE LIMITATIONS OF ADVANCED LLMs

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ABSTRACT

An exhaustive study was needed to test the limits of leading Large Language Models (LLMs) using numerous real-world clinical trial outcomes. It was also necessary to provide extensive hallucination studies based on both extracting main points and providing novel AI clinical decision support. Here, 100 Bevacizumab cancer therapy articles representing over 900K words were summarized by the 3.5 Sonnet model into 49K words, which completed a detailed and complex problem of several cancer and study types to press the capabilities of the ChatGPT o1 reasoning model. Report summaries in general followed an effective format, with guardrails to de-identify patient information and numerical data sources attributions to ground the output to the input. The main takeaway was that both LLMs typically remained in-context with the input data when more structured prompts were used, while precise quotations and author name citations were less prominent. These errors were likely due to LLM pressure towards achieving coherence vs. exact recall based on manufacturer inference-time compute settings. Overall, ChatGPT o1 provided state-of-the-art evidence-based Bevacizumab insights regarding clinical efficacy across indications, dosing recommendations, combination therapies, and biomarker-driven selections.

Keywords Clinical decision support - Bevacizumab clinical trials - Cancer - LLMs

1 Introduction

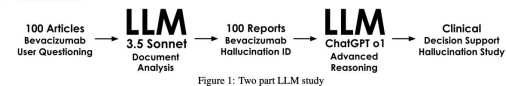


Figure 1: Two part LLM study

2025 LLM clinical decision support systems (CDSS) have pressed forward with application improvements based on newer AI models. In 2025, Buhr C., et al. provided decision-making assessments with local and web-based LLMs versus human board recommendations. ChatGPT 4o yielded a mean score of 4.7 (QR: 4-6), while Llama 3 [likely the 8B parameter model] achieved a score of 4.2 (QR: 3.5-5) [1]. Kim B., et al. proposed CDSS for mental health diagnosis that utilized GPT 4o, Gemini 1.5 Flash, and a Llama 3.2 model with constraint logic programming to translate diagnostic manuals to a logic program. GPT 4o had "9 correct diagnoses out of 10, followed by Gemini with 8 and Llama with 7" [2]. In addition, Ensina L., et al. utilized GPT 4o Turbo, 4, and 3.5 to analyze 969 electronic medical records in Brazilian and Portuguese regarding the detection of anaphylaxis. GPT 4 Turbo had the best performance, detecting anaphylaxis cases "with 90.6% precision, 100% sensitivity, 99.5% specificity, 99.5% accuracy, and a Cohen kappa coefficient of 0.95" [3].

Gao Y., et al. assessed instruction-tuned versions of Mistral 7B and Llama 3 70B models in predicting binary outcomes for sepsis, arrhythmia, and congestive heart failure; however a XGB classifier outperformed the smaller sized LLMs across all tasks [4]. In a *Nature Communications* article by Grist M et al. titled "Large Language Models lack essential metacognition for reliable medical reasoning", the authors found that GPT 4o, GPT 3.5 turbo, and several open source models typically did not possess "a robust capacity to provide higher accuracy". GPT 4o was found to be the best performing model, with a "missing answer recall metric" of $M = 46.1\%$, $SEM = 4.7\%$, which was significantly more accurate ($p < 0.0001$) than GPT 3.5 turbo's score of $M = 11.3\%$, $SEM = 2.9\%$ [5]. Steybe D., et al. used a retrieval-augmented generation (RAG) GPT 4 system "to provide answers on the prevention, diagnosis, and treatment of medication-related osteonecrosis of the jaw." Their GuideGPT framework using the Mann-Whitney U test showed significantly better ratings for GuideGPT than without using RAG "regarding content ($p = 0.006$), scientific explanation ($p = 0.032$), and agreement ($p = 0.008$), though not for language ($p = 0.407$)" [6].

100 preclinical/clinical/review papers: Bevacizumab multi-cancer

- Results: Key strategic combinations, biomarker patient selection.
- Advantages: 900K words to 490 word x 100 3.5 Sonnet reports.
- Limitations: Author performed initial 100 paper PRISMA search.

LLM Papers: Cancer Drug & Oncology Trials

COST CONTAINMENT OF GLOBAL MONOCLONAL ANTIBODY DRUGS AND CANCER CLINICAL TRIALS VIA LLM FOCUSED REASONING

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ABSTRACT

Expenses related to monoclonal antibody drugs worldwide and cancer clinical studies must be reduced in order to increase pharmaceutical industry efficiency. These financial opportunities can be assisted using state-of-the-art Large Language Models (LLMs) for focused report generation and advanced reasoning of solutions and forecasts that are based on authors' original findings. Here, Claude 3.5 Sonnet utilized 45 articles totaling approximately 357,000 words to effectively generate 45 reports. OpenAI ChatGPT o3-mini processed 15 of the reports to obtain comprehensive monoclonal antibody (mAb) cost solutions and financial forecasts. This included a financial recommendation of mAb biosimilars for a 55.2 percent price per dose decrease vs. a bevacizumab biologic due to Japan financial incentives, as reported by Itohima H. et al. The 30 additional reports were based on cancer clinical trial cost-effectiveness studies, with ChatGPT o3-mini reasoning to produce tables regarding economic strategies and projections. This included a "Total drug cost avoidance of \$92,662,609" over 10 years" when sponsored clinical trial participation was employed for solid tumors using various mAb therapies, as detailed by Carrera M. et al. 2024. The primary advantages of this comprehensive approach were: 1) Efficient report generations by 3.5 Sonnet of nearly 25,000 words in 20 minutes, 2) ChatGPT o3-mini's linear dependency prompt structure reduced narrative drift with structured outputs in less than 5 minutes, and 3) Ethical AI principles were strengthened: financial data was limited by rigorous prompts, yielding outputs that were highly traceable to source data using either LLM, as opposed to relying on the models' training data.

Keywords Cost-effectiveness · Monoclonal antibodies · Cancer clinical trials · LLM reasoning · Prompt engineering

1 Introduction



Figure 1: Simplified Report and Table Generations Process Diagram

The above process diagram represents the overall work flow in starting with complex and voluminous financial cancer drug articles, generating focused reports, and reasoning reports to yield practical and standardized information. Guardrails such as de-identification of patient information relevant to GDPR, AI Act and HIPAA compliance were implemented, as well as structured report guidelines, and limiting LLMs only to information derived from pdfs and subsequent text generations strengthened ethical AI principles. In addition, two different LLM manufacturers were used to reduce compound bias throughout the process. LLMs continue to be utilized by researchers in efforts to decrease costs associated with drug searches, drug pair interactions, clinical trial protocol generations, and clinical trial patient matching.

Li T. et al. in February 2024 proposed a new model using a few-shot learning approach to predict the synergy of drug pairs in tissues that lack structured data and features. Their experiments involved rare tissues from different cancer types to demonstrate that their CancerGPT model achieved significant accuracy using approximately 124M parameters. This accomplishment was comparable to a larger fine-tuned GPT-3 model with approximately 175B parameters. The authors claimed their breakthrough had significant implications for drug development which may be used in a cost-effective manner [1].

Chaves J. et al. of Google Research in June 2024 introduced Tx-LLM as a generalist LLM fine-tuned from PaLM-2 that "encodes knowledge about diverse therapeutic modalities", and was based on 709 datasets targeting "66 tasks spanning various

Monoclonal antibody 45 article input, 45 financial report output

- Results: ChatGPT o3-mini: 55.2% price per dose decrease (est.)
- Advantages: 10 yr Total drug cost avoidance of \$92,662,609 (est.)
- Limitations: SEK to US\$ conversion errors, Manual paper search.

AUTONOMOUS LLM AGENT AND SCALABLE REASONING LLM FOR GENERATING CANCER DRUG INDUSTRY COST SOLUTIONS

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ABSTRACT

Every once in a while, a new artificial intelligence technology is released that significantly improves research utility and results. This "deep research" tool released by OpenAI in February 2025 is a Large Language Model (LLM) web agent that autonomously queries sites such as PubMed Central (PMC), and generates high quality summaries with verifiable citations. A February 2025 article by Haman M. et al. reported deep research advantages in "analyzing 37 sources-35 of which were from the PubMed website" using the OpenAI o3 model. Here, ChatGPT 4.5 Deep research summaries regarding five pharmaceutical industry financial categories were found to be 100% in-context with PMC articles, and averaged 1,400 words in 10 minutes with minor issues. Also impressive was the processing of these summaries by the Claude 3.7 Sonnet Extended reasoning model to produce a structured 1,900 word 37 citation report containing detailed economic solutions, which were supported by 6 paragraphs of key insights collaborating multiple author quotations in approx. one minute. In addition, the Claude model produced eleven professional images based on USD or ROI trends, anomalies, and forecasts in three Python scripts. The Claude model possessed an output length that scaled by 3.2x for the report and 6x for code generations vs. the manufacturer's previous model, was 100% in-context with source data, and included interpretable reasoning summaries. The outputs from ChatGPT 4.5 Deep research served as inputs to 3.7 Sonnet Extended in mitigating model bias amplification that can occur when using results within a single software manufacturer. For transparency, comprehensive generation traceability analyses were conducted for the five summaries, the financial solutions report, and the eleven Python diagrams.

Keywords Autonomous LLM Agent · Reasoning LLM · Cancer · Drug Industry · Cost Solutions · Prompt Engineering

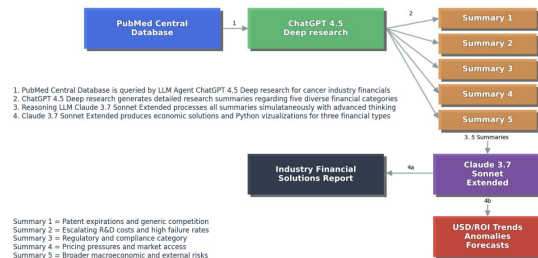


Figure 1: Main Process Diagram

PMC deep research cancer drug industry cost solutions

- Results: 6 Key insights on exclusivity, failed trials, biologics, etc.
- Advantages: 3.5 Sonnet Python charts: drug cost anomalies.
- Limitations: ChatGPT 4.5 Deep research 1 of 5 summaries re-run.

LLM Papers: Cancer Drug & Oncology Trials

END-TO-END PANCREATIC DUCTAL ADENOCARCINOMA DIGITAL TWIN CLINICAL TRIAL PROPOSALS

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Abstract

Question: Can online clinical trial literature be utilized by artificial intelligence to generate well-informed pancreatic ductal adenocarcinoma (PDAC) digital twin trial proposals?

Findings: AI successfully generated 40 meta-analyses in PDAC top research areas at an average word length of 10,196. One verification per meta-analysis was performed using AI automation, and the combined 408,081 word dataset was utilized to generate reports. Reports were separately verified and further visualized by AI. A second dataset consisting of the reports was used by five different AI models to yield five proposals. The proposals were evaluated by five AI judges to determine the top three proposals. Proposal deliverable completion reached a maximum score of 9.60/10, while the top citation proficiency score was 9.20. Prospective trial impact reached a peak score of 8.94, while funding probabilities ranged from 8.54 to 8.66.

Design: The framework consisted of 7 AI model combinations; o3re: ChatGPT o3 Research, g25p: Google Gemini 2.5 Pro Preview, son4: Sonnet 4 Extended, grk3: xAI Grok 3, o3pr: ChatGPT o3-pro, op4: Opus 4 Extended, and o3ch: ChatGPT o3; which were utilized according to model specific advantages. o3re autonomously searched the web and constructed detailed meta-analyses for Dataset 1. g25p was used to verify quantitative data across meta-analyses and reports, and also generated reports. son4 and op4 were used to produce visualizations in Python for reports and proposals; while grk3 fixed code when necessary. The 6 reports were combined to form Dataset 2, which was an input for the o3pr, op4, g25p, o3ch, and grk3 proposals. The same 5 models served as judges for the combined proposals, with average scores being used to determine the top 3 proposals. AI run time for core experiments was approximately 14 hours, with paper completion in 27 days.

Results: One verification for each meta-analysis was conducted by g25p at an average accuracy of 95%. The same model verified one table for each of the six reports at a self-reported 84.0% accuracy. Both verification accuracies are considered preliminary due to complexities in grounding multiple data types. Meta-analyses by o3re ranged from 6,440-13,033 words with an average time of 19.3 minutes. Reports by g25p averaged 2,538 words and 2.48 minutes, with a Pearson's coefficient of 0.094. Proposals totaled 8,903 words in 22.32 minutes. Proposal analyses by each of the five models were within 1-2 minutes, except for o3pr at 17.77 minutes. Each proposal included 10 prompt specific deliverables and a main therapy recommendation from a report.

Importance: An area of significance for all 5 proposals was the inclusion of a triple drug combination with the highest Therapeutic Synergy & Viability Score (TSVS) detailed in Report 1. Proposal A utilized the combination as follows: "The DT plan will prospectively simulate a first-in-human (FIH) study of the top-ranked therapeutic triplet—Daraxoxnasib + Mitazalimab + Liposomal Irinotecan—identified in Report 1 (predicted TSVS 8.15)." Proposal B also highlighted the therapy: "simulating 10,000 virtual pancreatic cancer patients to de-risk the clinical development of a novel three-drug combination: daraxoxnasib (pan-KRAS inhibitor) + mitazalimab (CD40 agonist) + liposomal irinotecan." The drug combination is traceable to source data using meta-analysis MA-23 for Daraxoxnasib, MA-15 for Mitazalimab, and a total of 8 Meta-Analyses utilizing Liposomal Irinotecan. This example of data fusion likely indicates functional utility between AI datasets.

Conclusion: Enhanced predictive performance and clinical relevance was achieved through a scalable multi-step workflow beginning with meta-analysis and report generations. Source data was utilized effectively across models, yielding AI aware PDAC digital twin trial proposals. Although automated AI verification accuracies were preliminary, reports were translated into top proposals, and screened visualizations were effective in discerning results. Vital trial proposal information obtained from charts revealed important distinctions between the three top proposals across 24 month timelines, multi-part ROI analyses, and budgets ranging from \$8.5M-\$17.8M. Proposal A by o3pr included the most intriguing digital twin Phase I/II trial, with high scores in deliverable completion, citation ability, and chances of funding.

CHATGPT 100,000 PATIENT 24-MONTH *In Silico* PHASE III 5-ARM PANCREATIC CANCER CLINICAL TRIAL TRIPPLICATE

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Abstract

Inquiry: Is it possible for ChatGPT to simulate three reproducible 100,000 patient pancreatic ductal adenocarcinoma (PDAC) Phase III clinical trial reports? If so, can the results be internally and externally validated, cross-verified using other AI models, and be compared both clinically and financially to other trials?

Concept: 5 arms based on the Daraxoxnasib + Mitazalimab + liposomal Irinotecan drug combination, baseline characteristics, and patient archetypes were identified from a prior study: doi.org/10.5281/zenodo.15735068. Six artificial intelligence models were then implemented to address the clinical trial pipeline: o3ph: ChatGPT o3-pro Research, g25p: Google Gemini 2.5 Pro, grk4: Grok 4, grk3: Grok 3 Think, o3pr: ChatGPT o3-pro, and op4: Opus 4 Extended. o3ph generated the ICH E3-aligned trial reports, log files, plus internal, and external validations. g25p, grk4, grk3, o3pr, and op4 provided cross verifications that highlighted trial-to-trial and model-to-model correlations. g25p utilized 24 generations in the study to produce a virtual trials overview, while o3ph provided a meta-analysis of pooled and scored data versus relevant virtual and on-site trials. o3ph also provided a financial assessment and value proposition of USD estimates against Phase II and Phase III studies. op4 provided visualizations written in Python for the majority of the sections.

Results: 100,000 individual patients generated from three separate o3ph conversations followed multiplicative hazard ratios and per-arm monthly hazards set in the prompt. Key variables were independent of each other, which yielded distributions of uncensored results. Log file cumulative effects of the censored 100,000 patients yielded expected results in OS by Arm (A > D > E), >G3 AE (A > D > E), and PFS (A > D > E). Baseline characteristics by metric across trials were in close alignment, and internal validations between log files and trial reports exhibited similar performance. External validation vs. a Flatiron Health dataset for OS passed, while ECOG validation saw higher differences. These deviations, along with a KRAS-mutant labeling issue were high, but consistent in magnitude across the three trials.

Outputs: In order to consolidate trial information, validations, and cross-verifications, g25p processed 24 of these relevant outputs to create a virtual trials overview. The core trial information, technical specifications, reproducibility, and validation findings provided a concise output needed for subsequent comparisons to trials. The method used to pool the current study with prior studies was accomplished by o3ph utilizing the virtual trials overview alongside online clinical trial data to produce a 9,574 word meta-analysis. Results focused on PRODIGE-4 and NAPOLI-1 trials that were top two in OS, while Arm A was third. However, the Arm D doublet of Daraxoxnasib + Mitazalimab was less toxic than the other trials, and was found to be more clinically feasible than FOLFIRINOX in PRODIGE-4.

Impacts: The financial assessment and value proposition performed by o3ph and visualized by op4 placed an estimated price of \$36,330 on the current study (1 user at \$150/hr working 60 hrs/wk). Estimates for other virtual trials ranged from \$120,000-\$600,000, while a real Phase II trial was \$20.0M, and the Phase III trial estimate was \$100.0M. Time-to-decision was fastest for the 100K Triplicate at 1 month while other studies ranged from 4.5 months to 5.0 years. The AI's main financial decision was that Arm A (Daraxoxnasib + Mitazalimab + liposomal Irinotecan) was not a strong enough candidate, and the results from the current study were estimated to save \$19,960M to avoid a clinical trial failure. In addition, a \$2.6M burn rate reduction was anticipated, with an overall cost reduction of 99.997% vs. a Phase III trial per patient.

Outcome: The main benefit that that reproducibility was observed across a single trial or multiple trials, while individual patients likely varied based on raw exponential sampling. The o3ph feat was primarily in providing a trial report that was replicable between the other trials performed in separate conversations. Similarly, the g25p model's processing of 24 outputs to create a virtual trials overview could not be accomplished by any other model due to token limitations. The overview served to inform the trial meta-analysis and financial assessment by o3ph, providing tangible comparisons and planning tools for upcoming studies. All work was performed by one user in a 30 day window.

40 PDAC area meta-analyses to 6 reports to Top 3 DT proposals

- Results: Budget projections: o3-pro \$17.8M, Opus 4 \$8.5M.
- Advantages: Daraxoxnasib selected by Gemini as top PDAC drug.
- Limitations: Gemini 2.5 Pro only for large, complex verifications.

100,000 patient PDAC Daraxoxnasib code-free simulation 3x

- Results: G3+ AE: Daraxoxnasib 8.0% over FOLFIRINOX 25.0%.
- Advantages: External validation, cross-verifications, value prop.
- Limitations: Trial to trial variability, KRAS G12C discrepancy.



LLM Papers: Cancer Drug & Oncology Trials

QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation From Protocol to Prediction: Code, VVUQ, and Playbook

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Abstract

Question: Is it plausible for artificial intelligence to generate QSP (quantitative systems pharmacology) pancreatic cancer clinical trial protocols, Python scripts, VVUQ (verification, validation, and uncertainty quantification), and playback?

Concepts: Two prior author studies consisting of drug arms, baseline characteristics, and patient archetypes were incorporated into the initial text trial protocol. The protocol was then further optimized and converted into Python by ChatGPT 5 Pro Research (ChatGPT). Additional trial attributes and mathematical functions were added; followed by hyperparameter optimizations, increased adverse events functionality, and fine-tuning of time steps and tumor grid size primarily by Gemini 2.5 Pro (Gemini). Sensitivity analyses to biological parameters were then conducted; yielding the final model code with optimized parameters and further comparisons to established trials including POLO, NAPOLI-1, and MPACT. The code was then converted back into a plain text protocol by ChatGPT for interpretability regarding non-technical staff, and serving as a platform for future developments.

Results: Verifications of several mechanistic variables revealed numerically stable objective response rates (ORR), with results assisting the finalization of time step $dt = 0.05$ and tumor volume grid size $= 5$. Uncertainty quantification of biological sensitive vs. resistant clone parameters also aided in robust ORRs for a combination therapy. Additional sensitivity analysis was performed regarding a KRAS inhibitor whose Emax potency increased by up to threefold, limiting ORR error to 11%. In effect, both numerical stability and biological credibility were demonstrated by the QSP pancreatic cancer simulation. Efforts to ground the control arms to external standards for this model had some success with mOS, ORR%, and DCR% performing more optimally; while mPFS, and Grade 3+ Adverse Events experienced less optimal results based on external validations.

Outputs: Initial text based protocols were iteratively refined by ChatGPT based on the author's prior empirical trial, known requirements for QSP trials, and additional biological incorporations. Code trials were generated based on iterative prompting to optimize parameters and variables. The 10 arm, 7 archetype, 10,000 patient trial with up to 250 ordinary differential equations (ODEs) per patient was aimed towards rapid drug prototyping a Phase II trial in real time. Each trial run yielded a Python script, notebook, and a patient log file. These generated materials were then used throughout the study to build the QSP playbook, VVUQ documentation, and visualization text instructions. The instructions were then converted into Python scripts by Opus 4.1 Extended (Opus), further optimized by the author, and then executed in Google Colab.

Impacts: Financial assessments were established between industry QSP virtual trial costs at \$2M vs historical Phase II and Phase III trials. The current study was performed by the author at a theoretical cost of \$36,304 based on a \$150/hr rate at 60 hr/wk for 4 weeks. A QSP cost reduction of up to 99.6% was found; supported by a 23 month time reduction and +27,500% ROI vs. a typical \$10.2M Phase II trial. Due to the larger patient cohorts and low resource requirements, the current QSP study was \$3.6/patient vs. an in-person trial of \$59,500/patient, a 16,528x difference. These financial propositions were most likely realized by preventing no-go arms from proceeding to in-person trials, such as Arm E; which had a low mOS at 6.6 month, a high Grade 3+ AEs at 92.5%, and second highest Drop % at 14.2%.

Outcome: Arm C, an oncogene-targeted therapy, was the most recommended drug combination by AI due to its high ORR at 70.8% and mOS as 11.9 (HR = 0.50). Arms H and I derived from the previous empirical trial were consistently top drug combinations due to competitive response rates and survival. AI models proved pivotal for specialized tasks as ChatGPT was the domain expert that automated text production, Gemini generated trial code solutions at scale, and Opus converted text instructions into effective Python dashboards. Ultimately, as trial code character numbers and complexity rose, models were able to address smaller challenges in the code to obtain useful updates. The numerical stability and sensitivity analyses throughout the VVUQ process can be considered among the study's most impactful results, while endpoints were more challenging to align with external control arms. This may be due to an estimated 2.5 million tumor volume states being updated at each time step, adding complexity for further AI modifications.

Accelerating FDA Compliance and Cost Efficiency of *in silico* Clinical Trials via AI Digital Twin Pancreatic Cancer Simulation

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Abstract

Question of Interest:

Can a bidirectional PDAC digital twin, subjected to M15-aligned verification, validation, uncertainty quantification, and applicability assessment, provide sufficiently credible comparative predictions of arm-level efficacy/safety (ORR, DCR, mPFS, mOS, HRs, G3+ AE, dropout) to prioritize Phase II platform-trial arms and inform design choices (e.g., eligibility mix, progression threshold, horizon), thereby reducing empirical iteration while preserving patient safety?

Context of Use:

A cohort-level digital twin simulates a 10-arm PDAC platform trial (~100 patients/arm, 36-month horizon, $dt=1$ day). Tumor dynamics use a two-compartment sensitive/resistant Emax framework with archetype-driven growth rates and a lognormal "sensitivity multiplier." Survival uses a base hazard with post-progression and biomarker multipliers; toxicity generates G3+ events with dropout probability; a rule-based policy effects $1L \rightarrow 2L \rightarrow BSC$ transitions. The twin executes a closed-loop sense-analyze-recommend-act-learn cycle with per-patient logs.

Data/knowledge used: drug parameters and archetypes transferred from prior QSP work and literature; control-arm external targets from MPACT (Arm A), NAPOLI-1 (Arm G), and POLO (Arms J/K). Specific role of model outcomes: rank arms, estimate HRs and endpoint distributions, test design assumptions. Other evidence: a priori VV40/PDA test suite executed-Verification/Numerics (V-01 de-convergence; V-02 seed reproducibility; V-03 zero-efficacy; V-04 zero-growth; V-05 boundary conditions; V-06 toxicity logic), Validation & Sensitivity (S-01, S-09, covering Emax/EC50/half-life, growth, resistance, dropout, hazard multipliers), UQ (UQ-01-.03, sigma variation, age SD, 10-seed ensembles), and Applicability (A-01 population drift, A-03 RECIST threshold, A-04 horizon; A-02 dosing schedule resilience). External validation: Arm A close on ORR and mOS, low on mPFS; Arm G underpredicts mPFS/OS; J/K further off-acknowledged calibration limitations.

Model Influence: Medium

Justification: The twin informs internal prioritization and design (arm ranking, sample size tuning, eligibility mix, sensitivity to rules) but is not the sole basis for regulatory or labeling decisions. Outputs are triangulated with literature comparators and expert judgment. Verification and UQ support reliable computation; partial external fit tempers influence.

Consequence of Wrong Decision: Medium

Justification: If mis-prioritized, resources could shift to a less active arm and expose Phase II participants to suboptimal regimens; however, care remains within accepted standards under IRB/DSMB oversight, and no patient-facing recommendations are made by the software. Thus, consequences are meaningful for efficacy and development efficiency, but with mitigations for safety.

Model Risk: Medium

Justification: Combining Medium Model Influence with Medium Consequence yields Medium risk (per M15 and ASME V&V 40 logic). Credibility activities were commensurate: code/calculational verification (dt stability; seed reproducibility; boundary logic checks), model robustness via sensitivity to key biological/clinical assumptions (S-01, S-09), stochastic/UQ ensembles (UQ-03), and applicability to population mix, RECIST thresholds, and horizon (A-01, A-03, A-04). External validation is partly met for A and G with documented gaps, transparently constraining scope.

Model Impact: Medium (with a path to High)

Justification: Relative to current regulatory expectations, the twin provides MIDD evidence suitable for planning and interaction: clearly stated Questions/CUs; risk-informed Model Evaluation; pre-specified Technical Criteria via the VV40 test suite; reproducible code and patient-level logs; ensemble uncertainty bands; and applicability analyses. This supports early FDA interactions (e.g., MIDDQ-sub) for arm prioritization and design what-if analyses, potentially saving time and cost versus purely empirical iteration. Impact is not rated High because simultaneous multi-endpoint external calibration (A, G, J/K) and prospectively defined quantitative acceptance targets/CIs need strengthening; completion of A-02 dosing applicability and expanded external validation would elevate impact.

PDAC Quantitative systems pharmacology, extensive VVUQ

- Results: Daraxonrasib drug triplet best mOS HR = 12.8 (0.25).
- Advantages: POLO, NAPOLI-1, and MPACT trial data, Python.
- Limitations: G3+ AE was consistently high for all patient arms.

Bidirectional PDAC DT simulation, treatment adaptations by time

- Results: Daraxonrasib drug triplet best mPFS HR = 7.5 (0.31).
- Advantages: 55 VVUQ tests for RECIST, Drug Params, Toxicity.
- Limitations: No patient-specific PK/PD, tumor growth parameters.

LLM Papers: Cancer Drug & Oncology Trials

End-to-End Oncology Clinical Trial LLM Efficiency For Industry Adoption with FDA/ICH Regulations

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ABSTRACT

The process of a new oncology treatment from drug discovery and preclinical studies through Phase III clinical trials and FDA review can take over a decade and cost hundreds of millions of dollars. Therefore, automated and easy to use large language models which have adapted to code generation for production use of sensitive data should be implemented into cancer trial workflows at scale. Here, LLMs were implemented throughout the study to create table templates, datasets based on author and online literature; and followed by triple and quintuple LLM workflows for optimization, table population, peer review, and meta-analyses. Primary models utilized were Sonnet 4.5 Extended, Gemini 2.5 Pro, GPT-5 High, and Grok-4 Fast - with Sonnet being used the most to quickly solve increasingly larger problems and reach consensus between other LLM responses. The iterated workflow features 12 tables from discovery through Phase III trials, NDA/BOA & FDA Review that identified potential prior LLM applications, with estimated LLM time and cost reductions for trial activities. These itemized values for each table were summed by AI and provided in a cost summary of an entire glioblastoma drug development process. The 2025 trial baseline total was estimated to be 10-15 years at a cost of \$328M-\$837M, while the LLM/AI assisted workflow was projected to be 4.8-7.7 years at \$215M-\$673M, which is 52-73% faster and 24-50% less expensive. Key performance indicators of baseline vs. LLM, acceleration and cost reduction areas, regulatory framework summary, and call to action were also provided by Sonnet, marking a transition to broader adoption of LLMs for oncology clinical trials. Note: LLM outputs and LLM-generated code locally executed with patient data may require FDA approval.

Keywords Oncology · Drug Discovery · Preclinical · Phase III Trial · FDA · LLM

Introduction

A baseline workflow in oncology clinical trials refers to the amount of time and money required to meet regulatory requirements in bringing a drug to market. This process entails drug discovery, preclinical, Phase I-III clinical trials, NDA/BLA, cross-phase management & MIDD to achieve final Food and Drug Administration (FDA) approval. AI estimates for a successful glioblastoma drug campaign is ~10-15 years at a total cost of ~\$282M-\$837M [1]; with phase trial failures escalating costs to \$1.5B-\$2.5B+. Therefore, LLMs should be used to increase efficiency in all relevant areas. In particular, Anthropic, Gemini, OpenAI, and Grok LLMs have become proficient in processing quantitative data with high contextual awareness in English; while effective LLM code generation can enable on-site processing of patient data [1, 2, 3, 4]. Developers have thus automated many tasks using LLMs at scale for drug synergy prediction by Li in 2024 [5]; and rational enzyme design with the E2 model by Arc Institute, Stanford University, and NVIDIA [6]. In addition, Sanofi is building biology foundation models to assist quantitative systems pharmacology (QSP) and digital twin simulations for Phase 1b trials and dose optimization [7].

Phase II trial patient enrollment could also be improved by the work of Beattie in 2024 with GPT-4 screening of over 200 patients in the n2c2 dataset [8]. In addition, Phase II trial data management & monitoring (21 CFR 11, ICH E6) could also see improvements, as shown by work of Shekhar [9] using nCODE profile generation with 92% accuracy. Kawchak in 2025 developed Python based PDAC QSP and digital twin clinical trial simulations [10, 11]. Phase III trials can similarly see a boost in efficiency regarding SAP finalization through the work of Kuo in 2025 using RAG-LLM and LoRA for SAP auto-population achieving 75% report drafting time reduction (p<0.01) [12]. In addition, SDTM programming & validation can see improvements based on Pharmaverse's [13] use of a sdtxml R package. Additionally, Pitzer has developed a natural language processing (NLP) platform for case processing of unstructured data [14], while Council for International Organizations of Medical Sciences (CIOMS) in 2025 reported on automated coding with 292M AEs processed annually [15]. Reason in 2024 published on a GPT-4 automated health economic model programming in R in (~12 min) per pNSCLC model [16]. Kawchak comprehensive VVUQ model validations with 55 tests addressed FDA M15 and ASME V&V 40 [10]. NDA/BLA regulatory submissions have been addressed by Narrativa with CTD Module 2 assembly and ICH M4 summaries [17], and a CSR Atlas Knowledge Graph for direct data extraction as submission-ready narratives with full traceability.

25+ Author, ChemicalQDevice prior LLM works into trial savings

- Results: Glioblastoma trial 52-73% faster, 24-50% cheaper (est.)
- Advantages: Existing LLM paper + trial activity → ΔTime ΔCost.
- Limitations: Many human prompts, additional LaTeX formatting.

LLM-Generated Glioblastoma Drug Synergy Machine Learning: From Rapid Code Prototypes to Project Deliverables Package

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ABSTRACT

Large language models (LLMs) are best utilized through their ability to rapidly prototype code which can be executed locally, causing minimal workflow disruption to existing drug development processes and addresses validation, verification, and transparency concerns. Essentially, LLM written code substituting human code mitigates concerns of LLM black box interpretability and explainability issues. Here, this study uses LLMs to develop, peer review, and make code corrections, yielding a feasible drug synergy machine learning pipeline. This was achieved utilizing Sonnet 4.5 Extended to provide a six machine learning model Python notebook and 1020x26 dataset of 153 unique glioblastoma drug interaction pairs in a single output, with ChatGPT 5 Thinking as a senior peer reviewer recommending fixes to label leakage, group-aware cross validation, and probability calibration. Subsequent Sonnet optimizations focused on an enhanced random forest model at an accuracy of 0.9804 and Macro-F1 of 0.9705 in predicting three drug synergy classes. The final Sonnet automated output bundle of notebook, summary, implementation guide, recommendations mapping, and README files ensured end-to-end reproducibility, auditability, and version control - mimicking a real-world ML release. Four non-Sonnet inference LLMs were used to provide exhaustive analyses regarding performance and industry relevance of the notebook and dataset. LLM code generation using widely available proprietary models has validation advantages over direct LLM prompting with reduced workflow complexity over fine-tuned, agentic, and augmented retrieval methods. Therefore: easy to use, cost effective, and mainstream LLMs proficient in web search and multi-format uploads are ideal for generating code that can be run locally for everyday drug development tasks; with direct LLM prompts for conducting AI peer review based on complex and voluminous data - marking the transition from LLM benchmark competitions to full reviewed LLM applications.

Keywords Cancer Drug Synergy · Glioblastoma · LLM · ML Code Generation

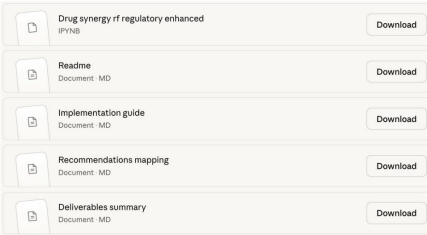


Figure 2B: Sonnet 4.5 Extended Single Output Regulatory-Enhanced Machine Learning Deliverables Ensures End-to-End Reproducibility, Auditability, and Completeness. Final Deliverables and Dataset

Glioblastoma drug synergy random forest ML deliverables

- Results: Three drug class = 0.9804 accuracy, 0.9705 Macro-F1.
- Advantages: Python notebook, 4 documents in a single prompt.
- Limitations: Only Claude Sonnet 4.5 Extended multi-file output.



LLM Papers: Cancer Drug & Oncology Trials

AI Peer Review Acceleration of LLM-Generated Glioblastoma Clinical Trial Patient Matching ML, FDA/ICH/ISO, and FastAPI

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ABSTRACT

Human peer review was an effective technique to address quality, originality, and errors of non-LLM generated work. Researcher-AI trust has since then grown due to performant large language model (LLM) benchmark improvements and author implementations across different model manufacturers for processing frequent, complex, and voluminous data. This single human-AI team has led to the emergence of readily available artificial intelligence peer review. A primary benefit of AI peer review is the ability to appropriately conduct iterative analyses and corrections throughout the entire manuscript process. This responsible method improves on over-worked and fatigued human recommendations provided after manuscript completion. Here, a maximum efficiency study was performed in 14 days with the author utilizing a prior glioblastoma clinical trial matching guidance and a TriAGPT paper as inputs to output a prompt for developing a Python deep learning pipeline. A 10 file dataset and clinical BERT model notebook were generated by Sonnet 4.5 Extended, followed by repetitions of code fixes and optimizations to yield a final notebook at a low test set performance of 67.3%. A triple AI peer review conducted by Sonnet, GPT 5.1, and Grok 4.1 all yielded the primary recommendation to upgrade to a tabular model, which was more suitable for the csv dataset.

ChatGPT then identified the most appropriate selection: an open source TabPFN-v2-clf model from Hugging Face. The Sonnet recommendations also included detailed instruction for five-fold cross-validation, SHAP explainability, and bias analysis incorporation. An AI peer review evaluation standard for ML workflows based criterion and efficiency metrics was generated by Opus 4.5 with highest efficiencies based on the least number of humans, prompts, and time at the lowest LLM cost. Subsequent evaluation of the current workflow with an overall quality of 87.5%, and 0.831 efficiency based on human, prompt, time, and cost metrics. The ChatGPT identified model and Sonnet review instructions were implemented into the prior notebook to yield a new notebook that saw an acceleration in research progress with a test set accuracy improvement to 94.0% with few-shot learning along with peer review corrections. A regulatory compliance prompt was created, followed by full document generations of FDA 21 CFR Part 820, ICH-GCP E6(R2), and ISO 14971 addressing quality, clinical practice, and risk management. Several attempts to create an effective FastAPI code repository by Sonnet were initially unsuccessful, however an Opus/Sonnet prompting strategy yielded a FastAPI interface with three successful glioblastoma patient clinical trial match predictions. AI as a senior software engineer and as a senior regulatory analyst were also employed as part of the AI peer review process.

The purpose of this AI peer review study was to place the world's focus back on innovating active and real world medical AI advancements in a fraction of the time. Measuring the effect AI implementation has on large workflows can be challenging to quantify. The conversational human-AI R&D relationship has become increasingly salient; therefore this single author project should be viewed as the standard for new workflows going into the future. This next evolution of peer review is especially suitable for hard working researchers in less favorable economic, educational, and affiliation conditions, such as the author of the paper. This process is afforded by low cost, fast, and easy to use state of the art LLMs from Anthropic, Google, OpenAI, and xAI, marking the pivot away from slow and irresponsible use of human peer review for LLM-generated data, and back to fast automation of patient health applications at scale. Many authors of less influence can now improve their speed of development and gain reputation in papers by simply providing prompts, LLMs, AI peer review recommendations, and corrections they made - without the need for paid review or journals. Interested parties can then view the author's peer review findings alongside the paper. Trust for the AI peer review process was established over a series of human-AI works reflected in LLM code and non-code benchmarking, applications in literature, and prior author works using AI in review roles coordinated across judging, external validation, cross-verification, meta-verification, and tests according to FDA/ICH guidance.

Keywords Clinical Trial Patient Matching · Glioblastoma · LLM Machine Learning Code Generation

GBM trial patient matching AI peer review, full FDA documents

- Results: 3x LLMs advice: 67.3% → 94.0% eligibility accuracy.
- Advantages: 6 FDA/ICH/ISO/HIPAA documents for compliance.
- Limitations: Opus 4.5, Sonnet 4.5 Extended ~1500 LOC limit.

Code Generation Competition: 16 Proprietary vs. Open-Source LLMs & Iterative Learning Based on FDA Adverse Event Reporting System

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ABSTRACT

Few effective goal-oriented iterative LLM code benchmarking studies exist. Successive high dimensional and complex problem improvements are desired versus conventional code assessments. Inspired by a recent CodeClast study, this tournament focuses primarily on the goal of generating functions to obtain a perfect competition task score based on three recent FDA FAERS files. Here, Opus 4.5 Extended was primarily utilized to build a novel Python evaluation engine measuring LLM code pair correctness, methodology, code quality, and algorithm effectiveness against a fixed reference standard and head-to-head. The notebook then automated Code A and Code B grading, and outputted their answers and reference standard of drug-reaction signals in csv files. The bracket was organized at scale: 16 LLMs - 8 proprietary LLMs on the left and 8 open-source LLMs on the right. The 8 Round 1 winners and corresponding notebooks were then re-introduced to each LLM with a competition prompt to generate the next round's code submission. Iterative learning in the form of improved final scores was observed for several Round 2 winners, which was based on its prior round competition code, competitors' code, and results. Gpt-5.2-pro and Gemini 2.5 Pro API were effective at iterative learning on the FAERS dataset goal; while Kimi K2 Thinking saw the biggest single round score increase at +0.405. Contestant models were from xAI, OpenAI, Gemini, Claude, DeepSeek, Kimi, GLM, MiniMax, and Qwen manufacturers.

Keywords Goal Oriented Software Engineering · LLM Code Generation · Iterative Learning · FDA FAERS

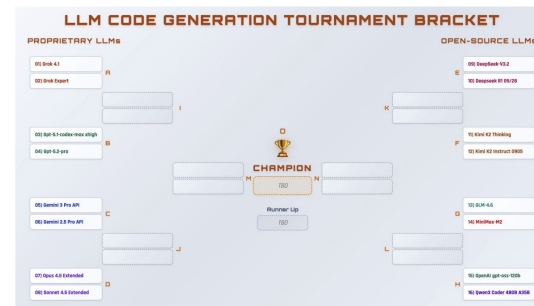


Figure 1: 16 LLM Single-Elimination Code Generation Bracket with Iterative Learning Process Throughout Rounds Based on FDA Adverse Event Reporting System (FAERS)

FDA Adverse event dataset, 16 LLMs, goal-oriented benchmark

- Results: Gpt-5.2-pro (0.9775) wins over gpt-oss-120b (0.2050).
- Advantages: Winners learned from previous round competition file.
- Limitations: No log files. Prior notebook competitor code, results.



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